

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_failure_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_195514_r
- started_at_local: 2026-05-13 19:55:14
- finished_at_local: 2026-05-13 19:58:09
- duration_hhmm: 00:03
- duration_seconds: 175.401
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.247848	0.012431	0.143218	0.834087	0.58	0.039893
cvrp_random_replay	random	score_only	False	0.244749	0.005376	0.138361	0.820371	0.58	0.035545
cvrp_failure_replay	failure	score_only	False	0.241217	0.005376	0.136399	0.8	0.78	0.022805
cvrp_failure_replay_compression	failure	novelty_gate	True	0.247998	0.005376	0.140166	0.842052	0.58	0.049725

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.247848, synthetic 0.012431, combined 0.143218.

- Accepted-epoch count 2, mean accepted code novelty 0.834087, and final complexity 0.58.
- Adaptation efficiency 0.039893 and archive sizes {'worst_cases': 5, 'failure_cases': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.247848 across 5 instances; family means: A=0.143979, B=0.254873, E=0.284069, P=0.301444.
- Panel synthetic_holdout mean gap 0.012431 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.028221.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.244749, synthetic 0.005376, combined 0.138361.
- Accepted-epoch count 2, mean accepted code novelty 0.820371, and final complexity 0.58.
- Adaptation efficiency 0.035545 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.244749 across 5 instances; family means: A=0.115183, B=0.259605, E=0.287908, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.241217, synthetic 0.005376, combined 0.136399.
- Accepted-epoch count 2, mean accepted code novelty 0.8, and final complexity 0.78.
- Adaptation efficiency 0.022805 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.241217 across 5 instances; family means: A=0.094241, B=0.241391, E=0.287908, P=0.341155.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 602, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.31441}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.247998, synthetic 0.005376, combined 0.140166.
- Accepted-epoch count 2, mean accepted code novelty 0.842052, and final complexity 0.58.
- Adaptation efficiency 0.049725 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.247998 across 5 instances; family means: A=0.115183, B=0.259605, E=0.287908, P=0.31769.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 602, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.31441}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of Replay-Aware CVRP Benchmark Results

Condition	Replay Mode	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Complexity	Code Novelty (mean)	Notes on Replay Mode
cvrp_no_replay	none	0.2478	0.0124	0.1432	0.58	0.8341	Baseline without replay
cvrp_random_replay	random	0.2447	**0.0054**	0.1384	0.58	0.8204	Improved transfer on synthetic holdout
cvrp_failure_replay	failure	**0.2412**	0.0054	**0.1364**	0.78	0.8000	Best transfer on CVRPLIB & overall
cvrp_failure_replay_compression	failure + compression	0.2480	0.0054	0.1402	0.58	0.8421	Compression pressure reduces transfer

Interpretations

- ****Optimality gaps on held-out CVRPLIB****:
- Best gap: ***cvrp_failure_replay*** (0.2412), followed closely by ***cvrp_random_replay*** (0.2447), both better than ***no_replay*** (0.2478).
- ***Failure_replay_compression*** is slightly worse (0.2480), likely due to compression pressure.
- ****Synthetic holdout gaps****:
- All replay conditions achieve lower synthetic gaps (0.0054–0.0124) compared to no replay (0.0124).
- ***cvrp_random_replay*** and ***cvrp_failure_replay*** tie with the best synthetic gaps (0.0054).
- Indicates improved generalization on synthetic instances by replay conditions.
- ****Transfer gap (combined measure)****:
- Lowest gap in ***cvrp_failure_replay*** (0.1364), then ***cvrp_random_replay*** (0.1384), both better than no replay (0.1432).

- Compression-aware replay (*failure_replay_compression*) again underperforms slightly (0.1402).
- **Complexity and Behavior**:
- *cvrp_failure_replay* has higher complexity (0.78) than others (0.58), potentially reflecting more refined behavior profiles (clustered_constructor).
- Others maintain balanced behaviors with lower complexity.
- **Code novelty**:
- Slightly decreases from *no_replay* (0.8341) to *failure_replay* (0.8000), despite improved transfer and optimality.
- Compression replay condition recovers novelty (0.8421) but at cost of transfer performance.
- Indicates reduced code novelty can coincide with increased transfer effectiveness.
- **Replay modes distinguished**:
- *No replay* baseline performs worst on transfer metrics.
- *Random replay* improves synthetic transfer gap best but not held-out CVRPLIB gap.
- *Failure replay* (focused on failure cases) yields best overall transfer, balancing held-out CVRPLIB and synthetic improvements.
- *Failure replay with compression* degrades transfer somewhat despite improved code novelty, showing compression pressure may limit beneficial adaptation.

Conservative Conclusions

- The **failure replay** condition clearly achieves the best transfer performance on held-out CVRPLIB instances and combined transfer metrics, despite having slightly lower code novelty and higher complexity. This suggests targeted replay of failures is beneficial for transfer.
- **Random replay** achieves the smallest synthetic holdout gaps but slightly underperforms on CVRPLIB held-out gaps, indicating that coverage of failure cases, not replay diversity alone, may be more critical for real benchmark transfer.
- Compression-aware replay reduces the effectiveness of transfer, indicating that replay compression pressure might inhibit adaptation useful for improving CVRP benchmarks.
- Higher code novelty does not necessarily translate to improved transfer; in fact, the best transfer result coincides with a moderate drop in code novelty.
- **No replay** lags behind all replay-based conditions on transfer metrics, confirming some form of replay helps generalization.

Recommendations

- Prioritize **failure replay** for improving transfer on realistic CVRP benchmarks.
- Avoid compression pressure in replay if transfer performance is paramount.

- Monitor complexity increase under failure replay but accept it given transfer gains.
- Interpret code novelty numbers cautiously; decreased novelty with improved transfer suggests novelty alone is insufficient to claim algorithmic invention.

This assessment relies primarily on optimality gaps on held-out CVRPLIB instances and synthetic holdout benchmarks to evaluate transfer quality, aligning with established evaluation criteria.